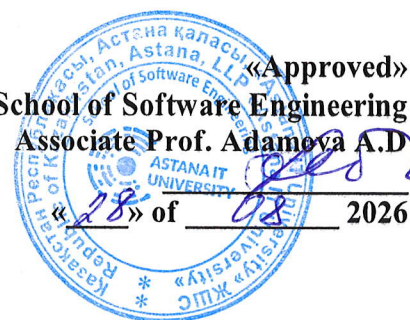




«Approved»
By Dean, School of Software Engineering
Associate Prof. Adamova A.D



Syllabus Academic Year 2026-2027

1. General Information																							
Course Code	SUBD 2217																						
Course title	Database management systems																						
Degree Cycle (Level)/ Major / Relation to curriculum	Bachelor's Degree 6B06101 - Computer Science 6B06102 - Software Engineering 6B06105 - Media Technologies 6B06301 - Cyber Security 6B04101 - IT Management 6B04102 - IT Entrepreneurship 6B06801 - Big Data in Healthcare																						
Year, trimester	2 nd year, 4 th trimester																						
Language of Instruction	English																						
Lecturer(s)/ Instructors/ Instructor Contact Information	Aivar Sakhipov, PhD in Computer Science, assistant professor, aivar.sakhipov@astanait.edu.kz, C1.3.354, office hours: Monday: 14.00-16.00 Abzal Kyzyrkanov, PhD in Computer Engineering and Software, assistant professor, kyzyrkanov.abzal@astanait.edu.kz, C1.3.354, office hours: Monday: 14.00-16.00 Samat Tankeyev, MSc in Computer Science and Engineering, lecturer, samat.tankeyev@astanait.edu.kz, C1.3.354, office hours: Monday: 14.00-16.00 Bekdaulet Shapigullin, MSc in Computer Science and Engineering, lecturer, bekdaulet.shapigullin@astanait.edu.kz, C1.3.354, office hours: Monday: 14.00-16.00 Bakgeldi Alkhabay, MSc in Computer Science and Engineering, lecturer, bakgeldi.alkhabay@astanait.edu.kz, C1.3.354, office hours: Monday: 14.00-16.00 Akzharkyn Zholmyrza, MSc in Informatics, lecturer, akzharkyn.zholmyrza@astanait.edu.kz, C1.3.360, office hours: Monday: 14.00-16.00 Miras Kabykenov, MSc in Computer Science and Engineering, lecturer, m.kabykenov@astanait.edu.kz, C1.3.357, office hours: Monday: 14.00-16.00																						
Number of credits	5																						
Workload of course components and credits per trimester	<table><tr><th rowspan="2">ECTS credits</th><th colspan="3">Contact hours</th><th rowspan="2">IASS</th><th rowspan="2">SIS</th><th rowspan="2">Total hours</th></tr><tr><th>Lectures</th><th>Practice sessions</th><th>Labor atory work</th></tr><tr><td>5</td><td>20</td><td>30</td><td>0</td><td>10</td><td>90</td><td>150</td></tr></table>						ECTS credits	Contact hours			IASS	SIS	Total hours	Lectures	Practice sessions	Labor atory work	5	20	30	0	10	90	150
ECTS credits	Contact hours			IASS	SIS	Total hours																	
	Lectures	Practice sessions	Labor atory work																				
5	20	30	0	10	90	150																	
Prerequisites	ICT or basic computer knowledge																						
Postrequisites	WEB Technologies, Capstone Project, Diploma Project																						
2. Goals, Objectives and Learning Outcomes of the Course																							
1. Course Description	The course “Database Management Systems” is a 10-week program focusing on concepts and structures necessary to design and implement a DBMS. It includes modern data models, database security, integrity, concurrency, query optimization, ACID properties, and normalization, with practical work in PostgreSQL.																						
2. Course Goal(s)	To introduce students to fundamental concepts and practices in database management																						

	systems and equip them with the skills to design, implement, and manage relational databases effectively. Course Objectives • Understand database fundamentals, types of DBMS, and their applications. • Design and interpret ER diagrams for database modeling. • Acquire skills in writing and executing SQL queries (DDL, DML, DQL). • Apply query optimization techniques. • Understand and implement ACID properties in transactions. • Apply normalization techniques to reduce redundancy. • Use PostgreSQL as a practical DBMS.
3. Course Learning Outcomes:	By the end of this course, students will be able to: • Design and implement databases using ER diagrams and normalization. • Write and optimize SQL queries. • Apply ACID properties in transactions. • Use PostgreSQL for practical database management. • Demonstrate problem-solving skills in database design and management.
4. Methods/forms of teaching	• Lectures and practices • Group work and case studies • Independent study (SIS) • Instructor-assisted student study (IASS)
5. Coursework and Grading Scheme	The final grade for this course will be based on a combination of individual assignments (6 assignments per the course), midterm, endterm and the final test. All grading follows the university's percentage scale. The formula for calculating the final grade: $0.3 \times 1st \text{ attestation} + 0.3 \times 2nd \text{ attestation} + 0.4 \times \text{Final}$. 1st attestation: Assignments: Assignment 1. Creating of ERD diagram – 20% Assignment 2. Intro to DDL – 20% Assignment 3. Intro to DML and JOIN operators – 20% Quizzes: Learn-quiz (on platform learn.astanait.kz) – 10% Midterm: Mixed format – 30% 2nd attestation: Assignments: Assignment 4. Subquery, grouping, aggregate, functions – 20% Assignment 5. Window functions – 20% Assignment 6. Set Operations, Grouping sets, Cube, and Rollup, PostgreSQL Views, PostgreSQL Indexes, ACID Properties in DBMS – 20% Quizzes: Learn-quiz (on platform learn.astanait.kz) – 10% Endterm: Mixed format – 30% Final Test – 100%
6. Academic Integrity	Students are expected to act with honesty and responsibility in their studies. Students are expected to do their own work, give proper credit when using others' ideas, and avoid cheating, plagiarism, or any form of dishonesty. Upholding these values ensures fairness and trust in the learning process for everyone.
7. Learning resources:	Main Literature: • Lecture slides (available on moodle.astanait.edu.kz), • Database Management Systems, 2023, Tulebayev Y. video lectures: learn.astanait.edu.kz • Ferrari Luca. Learn PostgreSQL [Текст] / L. Ferrari, P. Enrico. - 2nd ed. - UK : Packt Publishing, 2023. - 709 p. - ISBN 978-1-83763-564-1 : 41104.00. • Luzanov P. Postgres [Электронный ресурс] / P. Luzanov, E. Rogov, I. Levshin. - 11th ed. - 2025. - 187 p. - ISBN 978-5-6045970-8-8. • Kuzmiakova Adele. Creation and Management of Database Systems [Электронный ресурс] / A. Kuzmiakova. - Canada : Arcler Press, 2023. - 238 p. - ISBN 978-1-77469-673-6. • Shilds, W. (2023). SQL: Fast Immersion (in Russian). Progress Book, 224 p.

	• Jan L.Harrington. Relational Database Design and Implementation [Текст] / L.H. Jan. - 4 ed. - Amsterdam : Elsevier Inc., 2016. - 689p. Additional Literature: • M. McLaughlin, Oracle Database 12c PL/SQL Programming. New York, NY, USA: McGraw-Hill Education, 2014, ISBN: 978-0-07-181243-6. • Tulebayev Y.B., Sakhipov A.A., Database Management Systems. Astana IT University, 2026. – 143 p., ISBN: 978-601-12-8660-2 • Tulebayev Y.B., Sakhipov A.A., Database Programming PL/SQL. Astana IT University, 2026. – 109 p., ISBN: 978-601-12-8659-6
8. Technology employed:	• PostgreSQL, Oracle Live SQL • Moodle LMS, Microsoft Teams • Online SQL tutorials and emulators
9. Course Policies	Attendance: Attendance is mandatory. Students may not miss more than 30% of class sessions during the trimester. Exceeding this limit will result in an F (Fail) grade or may require the student to retake the course during summer school. This rule applies even if absences are supported by medical or other official documents. In cases of prolonged health issues or other serious circumstances, students must contact the university administration to formally discuss possible academic accommodations for the trimester or academic year. Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes. Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor. If a student is more than 5 minutes late for class or leaves without permission, his/her attendance will not be counted. Attestation I and II: Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course. Homework / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures. Assignment Defenses Policy. Assignment Defenses Requirement: Each student is required to defend their assignments according to the form provided by the instructor. Failure to defend assignments will result in a grade of zero for that particular work.

Evaluation Criteria: Marks for assignments will be awarded based on the quality of defense provided by the student. The defense process is integral to the assessment of the assignment.

Impact on Midterm and Endterm Exams: Assignments are designed to reinforce the understanding of course materials and prepare students for the midterm and endterm exams. If a student fails to defend any/all assignments, he may not be permitted to sit for the midterm or endterm exams. This is because the content and concepts covered in the exams are directly related to the assignments and their defenses.

Exceptional Circumstances: In cases of documented exceptional circumstances, such as medical emergencies or other valid reasons, students may be granted extensions or alternate arrangements for assignment defenses. However, these circumstances must be communicated to the instructor well in advance whenever possible.

Communication and Clarification: If students have any questions or concerns regarding the assignment defenses policy, they are encouraged to reach out to the instructor for clarification.

Note: It is the responsibility of each student to ensure they fulfill the requirements for assignment defenses in order to be eligible to sit for the midterm and endterm exams.

Late submissions: Most assignments will be discussed in class on the due date. All work must be submitted by the deadline; any submission **after the deadline is prohibited** and will automatically receive a grade of 0 (zero). All grading is based on a percentage scale.

In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.

Final exam : Final exam for the course "Database management systems" is conducted in the form of a Final Test. The test consists of theoretical and practical questions presented in a multiple-choice format. Upon completion of the test, students must submit their answers through the Learning Management System (Moodle) at moodle.astanait.edu.kz. Late submissions are not permitted.

Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor.

Online lessons can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround.

Cheating and plagiarism are defined in the Academic conduct policies of the university and include:

1. Submitting work that is not your own papers, assignments, or exams;
2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation;
3. Submitting or using falsified data;
4. Submitting the same work for credit in two courses without prior consent of both instructors.

Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university.

	<i>A report of the violation will be filed, and the student will be referred to the disciplinary committee for review.</i> Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email. Students should feel free to send an email if they have a question related to the course. The instructor responds as soon as they can but not always instantaneously. Please note that communication with the instructor is available during working hours, from 9:00 AM to 6:00 PM. Additionally, students are welcome to arrange a one-on-one meeting with the instructor at their office during office hours to discuss the class, either offline or online.
10. Course Schedule	• Weeks 1–2: Database fundamentals, ER modeling, normalization. • Weeks 3–5: SQL (DDL, DML, queries, joins), midterm exams. • Weeks 6–8: Subqueries, aggregation, window and set operations. • Weeks 9–10: Views, indexes, transactions, triggers; endterm exams.
11. Syllabus Inclusion Statements	• Accessibility: All course resources (presentations, databases, practice instructions) are available on Moodle. Students may request alternative formats of assignments if needed. • Academic Support: Instructor consultations and example SQL/PLpgSQL codes will be provided through Moodle to support independent practice. • Use of Technology and AI: Students are encouraged to use professional tools such as Oracle Live SQL and PostgreSQL emulators. Generative AI (e.g., ChatGPT) may be used only as a reference for examples but cannot replace independent work. • Course Community: Success in this course requires regular practice. Students are encouraged to collaborate in small groups, share ER-diagrams, discuss SQL queries, and exchange solutions as part of active learning.

3. Course Content

3.1 Abbreviations

#	Abbreviation	Meaning
1	IASS	Instructor assisted student study - Instructors assist students in achieving success by delivering support, feedback, and resources. This support might take the form of clarifying difficult topics, recommending additional materials, or providing individualized guidance.
2	SIS	Student independent study
3	IW	Individual work
4	GW	Group work
5	PA	Practical assignment
6	MCQ	Multiple choice quiz
7	PBT	Paper-based test

3.2 Course Plan

Week	Course Topic	Lectures (h/w)	Practice (h/w)	Lab (h/w)	IASS (h/w)	SIS (h/w)
1	Introduction to DBMS • What is a database? Properties and characteristics. • Database Management Systems: applications and examples. • Types of database models: advantages and disadvantages. • Entity-Relationship (ER) diagrams: entities, attributes, relationships.	2	3	0	1	9
2	Attributes in DBMS & Normalization • Attributes in DBMS: types of attributes.	2	3	0	1	9

	- Keys: super key, primary key, foreign key. - Functional dependencies: types and role in design. - Normalization: 1NF, 2NF, 3NF, BCNF.					
3	SQL & Data Definition Language (DDL) - Introduction to SQL: history, advantages, types. - Data Definition Language: managing tables. - Constraints: primary key, foreign key, unique, not null.	2	3	0	1	9
4	Data Manipulation Language (DML) & Data Query Language (DQL) - Data types in PostgreSQL. - Data Manipulation Language: INSERT, UPDATE, DELETE. - Data Query Language: SELECT. - String functions, filtering data, conditional expressions, operators.	2	3	0	1	9
5	PostgreSQL Joins Joins in SQL: inner join, full join, left join, right join, cross join, natural join, self join.	2	3	0	1	9
6	Grouping Data & Subqueries - GROUP BY clause. - Single-row and multi-row subqueries.	2	3	0	1	9
7	Aggregate & Window Functions - Aggregate functions: SUM, MAX, MIN, COUNT, AVG. - Window functions: RANK, DENSE_RANK, ROW_NUMBER, NTILE.	2	3	0	1	9
8	Set Operations & Grouping Sets - UNION, UNION ALL, INTERSECT, EXCEPT. - Advanced grouping: CUBE and ROLLUP.	2	3	0	1	9
9	PostgreSQL Views & Indexes - Creating and managing views. - Updatable and materialized views. - Indexes: unique, multicolumn, reindex.	2	3	0	1	9
10	Transactions & Triggers - Database transactions: ACID properties. - Row-level and statement-level triggers.	2	3	0	1	9
	Total:	20	30	0	10	90

3.3 Detailed course plan

Week No	Lecture Topics with Learning Resources	Tasks and Activities for Practice Sessions with Recommended Reading Resources	Practice Exercises, Assigned Activities, and Supporting Reading Resources
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1	Introduction to DBMS • What is a database? Properties and characteristics. • Database Management Systems: applications and examples. • Types of database models: advantages and disadvantages. • Entity-Relationship (ER) diagrams: entities, attributes, relationships.	Practice ER modeling with examples. Reading: Lecture slides (available on moodle.astanait.edu.kz), Ferrari Luca. Learn PostgreSQL (Ch.1–2).	Practice: Create ER diagram for a simple system (e.g., student database). Resources: PostgreSQL, Moodle materials.
2	Attributes in DBMS & Normalization • Attributes in DBMS: types of attributes. • Keys: super key, primary key, foreign key. • Functional dependencies: types and role in design. • Normalization: 1NF, 2NF, 3NF, BCNF.	Practice normalization tasks with sample relations. Reading: Lecture slides (available on moodle.astanait.edu.kz), Luzanov P. Postgres	Practice: Normalize given tables up to 3NF. Resource: PostgreSQL schema examples.
3	SQL & Data Definition Language (DDL) • Introduction to SQL: history, advantages, types. • Data Definition Language: managing tables. • Constraints: primary key, foreign key, unique, not null.	Lecture slides (available on moodle.astanait.edu.kz), Practice writing DDL queries (CREATE, ALTER, DROP). Reading: Kuzmiakova A.. Creation and Management of Database Systems (Ch.2–3).	Practice: Implement tables with constraints. Resource: PostgreSQL.
4	Data Manipulation Language (DML) & Data Query Language (DQL) • Data types in PostgreSQL. • Data Manipulation Language: INSERT, UPDATE, DELETE. • Data Query Language: SELECT. • String functions, filtering data, conditional expressions, operators.	Practice SELECT queries with conditions and operators. Reading: Luzanov P. Postgres Lecture slides (available on moodle.astanait.edu.kz)	Practice: Write queries using WHERE, ORDER BY, LIKE, BETWEEN.
5	PostgreSQL Joins • Joins in SQL: inner join, full join, left join, right join, cross join, natural join, self join.	Practice queries with different join types. Reading: Tulebayev Y., Database Management Systems Lecture slides (available on moodle.astanait.edu.kz)	Practice: Implement join queries on multi-table database. Midterm exam (Mixed format).
6	Grouping Data & Subqueries • GROUP BY clause. • Single-row and multi-row subqueries.	Practice subqueries and grouping tasks. Reading: Luzanov P. Postgres Lecture slides (available on moodle.astanait.edu.kz)	Practice: Write queries with GROUP BY and nested SELECT.
7	Aggregate & Window Functions • Aggregate functions: SUM, MAX, MIN, COUNT, AVG. • Window functions: RANK, DENSE_RANK, ROW_NUMBER, NTILE.	Practice aggregate and ranking functions. Reading: Tulebayev Y., Database Management Systems Lecture slides (available on moodle.astanait.edu.kz)	Practice: Write queries with window functions in PostgreSQL.
8	Set Operations & Grouping Sets • UNION, UNION ALL, INTERSECT, EXCEPT. • Advanced grouping: CUBE and ROLLUP.	Practice set operations on sample databases. Reading: - Jan L.Harrington. Relational Database Design and Implementation Lecture slides (available on moodle.astanait.edu.kz)	Practice: Apply CUBE and ROLLUP queries.

9	PostgreSQL Views & Indexes • Creating and managing views. • Updatable and materialized views. • Indexes: unique, multicolumn, reindex.	Practice designing and querying views. Reading: Ferrari Luca. Learn PostgreSQL Lecture slides (available on moodle.astanait.edu.kz)	Practice: Create and optimize views and indexes.
10	Transactions & Triggers • Database transactions: ACID properties. • Row-level and statement-level triggers.	Practice transaction control (COMMIT, ROLLBACK). Reading: Jan L.Harrington. Relational Database Design and Implementation Lecture slides (available on moodle.astanait.edu.kz)	Practice: Implement triggers in PostgreSQL. Endterm exam (Mixed format).

3.4 List of assignments for Student Independent Study

№	Assignments (topics) for Independent Study	Hours	Recommended Literature and Other Sources	Form of Submission
1	Fundamentals of DBMS: What is DBMS, database types, use cases, ER diagrams, cardinality	9	Reading list, resources	Electronic version (Moodle)
2	Keys and Normalization: super key, candidate key, primary/foreign keys; Functional dependencies; Normal forms (1NF, 2NF, 3NF, BCNF)	9	Reading list, resources	Electronic version (Moodle)
3	SQL DDL: CREATE, ALTER, DROP, TRUNCATE; constraints (PK, FK, UNIQUE, NOT NULL)	9	Reading list, resources	SQL script uploaded to Moodle
4	SQL Queries: SELECT statements, column aliases, ORDER BY, DISTINCT, data types, built-in functions, WHERE, LIMIT, FETCH, IN, BETWEEN, LIKE, IS NULL, logical operators, CASE	9	Reading list, resources	SQL queries in Moodle
5	SQL Joins: INNER, LEFT, RIGHT, FULL, CROSS, SELF, NATURAL JOIN	9	Reading list, resources	SQL queries (Week 5, before Midterm)
6	Subqueries and Grouping: GROUP BY, HAVING, nested SELECT, EXISTS, ANY, ALL	9	Reading list, resources	SQL solutions uploaded to Moodle
7	Aggregate & Window Functions: SUM, COUNT, AVG, MIN, MAX, RANK, DENSE_RANK, ROW_NUMBER, NTILE	9	Reading list, resources	SQL queries in Moodle
8	Set Operations: UNION, INTERSECT, EXCEPT; Grouping sets, CUBE, ROLLUP	9	Reading list, resources	SQL queries uploaded to Moodle
9	Views & Indexes: creating and managing views, materialized views, unique and multicolumn indexes, reindex	9	Reading list, resources	SQL tasks in Moodle
10	Transactions & Triggers: ACID properties, COMMIT, ROLLBACK, row-level and statement-level triggers	9	Reading list, resources	SQL report (Week 10, before Endterm)

3.5 For instructor assisted student study (IASS) schedule and policy follow your course LMS Moodle.

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Form of assessment	Total
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1 st attestation	Assignments: Assignment 1. Creating of ERD diagram Assignment 2. Intro to DDL Assignment 3. Intro to DML and JOIN operators Learn - quiz (on platform learn astana.kz) Mid Term*	20 20 20 10 30	Practical task, oral defense, in practice session time, quizzes, submission in https://lms.astana.kz	100
2 nd attestation	Assignments: Assignment 4. Subquery, grouping, aggregate, functions Assignment 5. Window functions Assignment 6. Set Operations, Grouping sets, Cube, and Rollup, PostgreSQL Views, PostgreSQL Indexes, ACID Properties in DBMS Learn- quiz (on platform learn astana.kz) End Term*	20 20 20 10 30	Practical task, oral defense, in practice session time, quizzes, submission in https://lms.astana.kz	100
Final exam*	Final Test. (MCQ)		theoretical and practical questions presented in a multiple-choice format in https://lms.astana.kz	100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final			100

* Final exam will be declared in MCQ format by practice instructor.

* MidTerm, EndTerm will be declared in Mixed format (Multiple choice questions and written question components) by practice instructor.

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	
B	3,0	80-84	Good
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	
C-	1,67	60-64	Satisfactory
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	
F	0	0-24	Fail

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Learning Outcomes	Indicators	Excellent (90–100)	Good (70–89)	Satisfactory (50–69)	Fail (0–49)
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1. Design and interpret ER diagrams	Ability to create ER-diagrams with correct entities, attributes, and relationships	Creates fully correct and complex ER-diagrams with clear cardinality and normalization applied	Creates mostly correct ER-diagrams with minor inaccuracies	Creates basic ER-diagrams but with missing relationships or normalization errors	Unable to create meaningful ER-diagrams
2. Apply normalization techniques (1NF–BCNF)	Correct use of functional dependencies and keys	Fully normalizes database up to BCNF, clearly explains functional dependencies	Normalizes database up to 3NF with small mistakes	Demonstrates only basic knowledge of 1NF–2NF, fails to justify higher normal forms	Cannot apply normalization, confuses keys and dependencies
3. Write SQL DDL and DML queries	CREATE, ALTER, INSERT, UPDATE, DELETE	Writes syntactically correct and optimized queries covering all requirements	Writes correct queries with minor inefficiencies	Writes queries with limited functionality or syntax errors	Cannot write correct queries
4. Write and optimize SQL SELECT queries with filtering and joins	Use of WHERE, ORDER BY, functions, joins	Creates complex SELECT queries using multiple joins, filtering, and functions accurately	Writes correct queries but with limited complexity or optimization	Produces only simple queries with frequent errors in joins/filters	Fails to write correct SELECT queries
5. Use advanced SQL (subqueries, aggregation, window functions, set operations)	GROUP BY, HAVING, nested SELECT, RANK, UNION, etc.	Correctly applies advanced SQL constructs with optimization and explains results	Applies advanced constructs but with some inaccuracies	Uses only basic queries with limited advanced features	Cannot demonstrate understanding of advanced SQL
6. Implement views and indexes in PostgreSQL	Creating and managing optimization objects	Creates efficient views and indexes, explains performance impact	Creates views/indexes with minor errors in logic	Creates simple views but does not understand performance aspects	Fails to implement views or indexes
7. Apply ACID properties in transactions and implement triggers	Transaction management and automation	Implements transactions with full ACID compliance; creates working row-level and statement-level triggers	Implements transactions and triggers with small logical errors	Implements only simple transactions, limited knowledge of triggers	No understanding of transactions or triggers

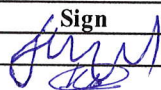
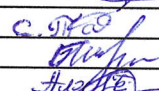
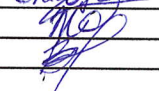
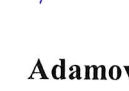
5. Course assessment methodology

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, practice works' defense and in-class assessments.

- **IASS (Instructor assisted student study)** - Instructors assist students in achieving success by delivering support, feedback, and resources. This support might take the form of clarifying difficult topics, recommending additional materials, or providing individualized guidance.
- **SIS** – Student engagement in self-directed study on assigned topics and the completion of relevant assignments given in the syllabus.
- **Mid-term and End-term** are a review of the topics covered and an assessment of each student's knowledge. The form of the midterm and end term exams will be declared in Mixed format (Multiple choice questions and written question components) by practice instructor.

- **Final assessment** for the course “Database management systems” is conducted in the form of a Final Test. The test consists of theoretical and practical questions presented in a multiple-choice format. Upon completion of the test, students must submit their answers through the Learning Management System (Moodle) at moodle.astanait.edu.kz. Late submissions are not permitted.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Aivar Sakhipov	Assistant professor	28/08/2026	
Abzal Kyzyrkanov	Assistant professor	28/08/2026	
Samat Tankeyev	Lecturer	28/08/2026	
Bekdaulet Shapigullin	Lecturer	28/08/2026	
Bakgeldi Alkhabay	Lecturer	28/08/2026	
Akzharkyn Zholmyrza	Lecturer	28/08/2026	
Miras Kabykenov	Lecturer	28/08/2026	

**Dean of School of
Software Engineering**

Adamova A.D.